

SI 544 - Introduction to Statistics and Data Analysis

Fall 2025: Thursday 1:30 - 3:30 PM, 1125 LCSIB

(All times are Ann Arbor local time.)

Instructor: Dr. Aasakiran “Bobby” Madamanchi

GSIs (alphabetical order):

David Ruttenberg (drutt@umich.edu) - Sections 6 and 7 (Tuesdays 6:30-7:30pm and 7:30-8:30pm, 2270 LCSIB)

Jackie Hu (yuuhu@umich.edu) - Section 4 and 5 (Wednesdays 4:30-5:30pm and 5:30-6:30pm, 2270 LCSIB or via Zoom at <https://umich.zoom.us/j/96182795707>)

Mohit Kumar (mohittk@umich.edu) - Section 1 and 2 (Mondays 1:30-2:30pm and 2:30-3:30pm, 2270 LCSIB)

Office hours

Bobby Madamanchi – by appointment over Zoom - All day Wednesday and after class Thursday are allotted for my office hours
<https://calendly.com/amadaman/umsi-office-hours>

David - by appointment over Zoom <https://calendly.com/drutt-umich>

Jackie - on Wednesdays:

in person at Leinweber 1228 at 3:30-4:30pm or

by appointment over Zoom

<https://calendar.app.google/GK9bibvRuGVvdb446>

Mohit - on Fridays:

by appointment over Zoom <https://calendly.com/mohittk-umich/15min>

Communication policy

This class uses Piazza for asking and answering course content questions. Email can be used for scheduling meetings and/or discussing unusual circumstances. Things related to grading/assignments should be emailed to your GSI and me (amadaman@umich.edu)

You can expect a response within 48 hours during normal business days (Mondays to Fridays). Communications that aren't shared in the proper location may take longer to receive a response.

Course Description

This course introduces students to statistics as a way of thinking about the world. Students will develop a vocabulary, conceptual framework, and methodological toolkit for describing and drawing inferences about datasets. We apply core statistical techniques to reason critically about real-world problems drawn from a variety of disciplines (sports, business, music, economics, science, and public policy). Students develop competency in using the R statistical programming language and in reading statistical analyses. No prior coding experience is required.

Learning Objectives

By the end of the course, we hope you'll achieve several learning objectives:

Competency. You will be able to independently perform statistical analyses and programming involving:

- Importing and tidying data
- Data visualization
- Summary statistics
- Probability
- Simple linear regression
- Linear regression using categorical variables

Literacy. You will know enough about the following concepts to play a meaningful role in a team doing statistical analysis:

- Confidence intervals
- Hypothesis testing
- Regression diagnostics

- Interactions
- Multiple regression
- Using the R programming language

Awareness. You will be aware that these concepts exist in statistics and have enough context and confidence to develop your knowledge further if needed in your career:

- Confidence Bands
- Prediction Intervals

Schedule/Readings

Course content is delivered and organized through the modules in Cavnas. The points and deadlines in the modules are the authoritative source. A tentative schedule for the semester is as follows, and HWs and Project Deliverable submissions will be in Canvas Modules

Week	Discussion	Lecture Date	Topic	Quizzes
1	No Discussion	8/28	Intro to Course Why Statistics	
2	Install RStudio	9/4	Descriptive Stats & Probability	Quiz #1: Syllabus & Descriptive Stats
3	Descriptive Stats & Probability	9/11	Data Wrangling & Visualization	Quiz #2: Visualization Quiz #3: Data wrangling
4	Exam 1	9/18	Sampling	
5	Sampling	9/25	Bootstrapping & Confidence Intervals	
6	Bootstrapping	10/2	Hypothesis Testing	Quiz #4: Hypothesis testing & sampling
7	Review	10/9	Exam 2	
BYE	No Discussion	10/16	NO CLASS	

9	Project Workshop	10/23	Regression	Quiz #5 Regression
10	Regression	10/30	Multiple Regression	Quiz #6: Multiple regression
11	Multiple Regression	11/6	Inference for Regression	Quiz #7: Inference
12	Review	11/13	Exam 3	
13	Project Help	11/20	Advanced Topics	
BYE	No Discussion	11/27	NO CLASS	
15	Project Help	12/4	Project Help	

Computers & software requirements

You will need a laptop or access to a computer. If you are having trouble getting access to a working computer, please contact the teaching team.

Class format

We meet once a week in lecture on Thursdays and once a week in discussion sections. During the Thursday sessions, activities will include live coding, question-and-answer sessions, and time dedicated to applying principles learned from readings to real-world data. These sessions will be recorded and made available shortly after the class ends for students who are not able to attend in-person.

Discussion Sections

Discussion sections are designed to give you additional opportunities to learn about applying the tools and techniques covered in the lecture material and synchronous sessions to novel datasets and scenarios. You will meet weekly with approximately 25 other students and your GSI.

There are 10 discussion section problem sets available, and we will automatically drop your two lowest grades. Your total discussion section score in the class will be

determined from your best 8 grades. Discussion section work must be turned in by the date in the assignment deliverable on canvas - typically the end of the following week.

Attendance and remote learning policy

This class is fully in-person and remote learning is supported for only special cases that are negotiated with the instructional team beforehand. We do not take attendance but your participation is expected.

Assignments & Assessments

Reading quizzes

There are readings assigned for many weeks. These are supplements to the lectures. You will get much more out of class if you read these before you attend. To encourage you to complete the readings before class, some weeks include a Canvas quiz.

You can attempt every quiz twice and you should use the week's reading as a resource.

Note that there will be more points available via quizzes than you can accumulate. You can collect a maximum of 100 points, but there are more than 100 points available from quizzes.

Exams

There are 3 exams in this course for 200 points each. They assess your understanding of material covered in lecture and the textbook. These are pen-and-paper assessments that do not require a calculator or computer.

Some exam questions will be based on problems from your discussion section worksheets and in-class lectures.

Final project - Individual

In lieu of a final exam there is a creative final project worth 200 points towards your final grade. You'll get to choose your own dataset to analyze, so this project can be relevant to your own research interests. The final project is worth 200 points and will be due on the last day of classes. More details about the assignment will be provided during the course.

[Places to look for data](#)

Grading

Discussion section work (drop 2 lowest): 300 points

Exams (3) : 600 points

Quizzes (drop 2 lowest): 100 points

Final project: 200 points

You will be graded out of a total of 1000 points. Note that there are 1200 points available.

The mapping to final grades is:

950-1000	A+
900-950	A
850-900	B+
800-850	B
750-700	C+
700-750	C
650-700	D+
600-650	D
<600	F

Requests for re-grading

We only regrade exams. The excess points available in the course provides leeway to makeup for irregularities in grading of assignments. Please email with any questions about exam grades.

Late Policy

You have ten (10) free late days to use during SI 544. Late days are not applicable to pre-lecture reading quizzes. One late day equals exactly one 24-hour period after the due date of the assignment (including weekends). No fractional late days: they are all or nothing. Once you have used up your late days, there will be a **25% penalty** for each subsequent 24h period after the deadline that an assignment is late. This is automatically applied on Canvas.

You don't need to explain or get permission to use late days, and we will track them for you. Note that resubmissions after the deadline will be counted as late submissions.

Academic honesty & collaboration policy

Learning statistical methods is often challenging. We are going to cover a wide range of topics in the course and we will move quickly between topics. Because it is our goal for you to succeed in the course, we encourage you to get help from anyone you like. Teamwork is also an essential part of the practice of statistics in nearly every domain.

As a matter of academic integrity, the final submission of each exercise must be in your own words. If you receive assistance on an assignment, please indicate the nature and the amount of assistance you received. If the assignment is computer code, add a comment indicating who helped you and how. Any excerpts from the work of others must be clearly identified as a quotation, and a proper citation provided (e.g., in the comments of the code if it is a code fragment you have borrowed).

If you are a more advanced student and are willing to help other students, please feel free to do so. Just remember that your goal is to help teach the material to the student receiving the help. It is acceptable for this class to ask for and provide help on an assignment via Piazza, *including posting (short!) code fragments*. Just don't post complete answers. If it seems like you've posted too much, one of the instructional staff will contact you to let you know, so don't worry about it. When in doubt, err on the side of helping your fellow students.

To reiterate, the collaboration policy is as follows: Collaboration in the class is allowed (and even encouraged) for assignments – you can get help from anyone as long as it is clearly acknowledged.

ChatGPT policy

You are welcome to use ChatGPT and other generative AI tools on all assignments except the in-class exams. You must cite these tools when you turn in your work, though. You must also take full responsibility for your submissions—any coding or conceptual errors will be graded as your own, regardless of whether they came from ChatGPT. In other words: proofread everything that comes out of generative AI before you turn it in.

You may find it simpler to complete the assignments using your own knowledge and expertise rather than AI tools. It will also be challenging to complete the in-class exams if you are reliant on AI.

Plagiarism policy

The instructors actively check for plagiarism and have zero tolerance. If caught, you will receive a zero for that portion of the assignment (or the whole assignment), depending on the scope of the plagiarism. All cases will be referred to the Office of Academic Integrity regardless of their scope.

UMSI Statement on Academic Integrity

Unless otherwise specified in an assignment all submitted work must be your own, original work. Any excerpts, statements, or phrases from the work of others must be clearly identified as a quotation, and a proper citation provided. Any violation of the School's policy on Academic and Professional Integrity (stated in the Master's and Doctoral Student Handbooks) will result in serious penalties, which might range from failing an assignment, to failing a course, to being expelled from the program. Violations of academic and professional integrity will be reported to UMSI Student Affairs. Consequences impacting assignment or course grades are determined by the faculty instructor; additional sanctions may be imposed by the assistant dean for academic and student affairs.

Accommodations for Students with Disabilities

The University of Michigan recognizes disability as an integral part of diversity and is committed to creating an inclusive and equitable educational environment for students with disabilities. Students who are experiencing a disability-related barrier should contact [Services for Students with Disabilities](https://ssd.umich.edu/) (https://ssd.umich.edu/; 734-763-3000 or ssdoffice@umich.edu). For students who are connected with SSD, accommodation requests can be made in Accommodate. If you have any questions or concerns please

contact your SSD Coordinator or visit SSD's Current Student webpage. SSD considers aspects of the course design, course learning objects and the individual academic and course barriers experienced by the student. Further conversation with SSD, instructors, and the student may be warranted to ensure an accessible course experience. The instructional team will treat any information that you provide in as confidential a manner as possible.

If you need to speak privately about an accommodation request: please email me at amadaman@umich.edu

Student Mental Health and Wellbeing

University Students may experience stressors that can impact both their academic experience and their personal well-being. These may include academic pressures and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc. If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. Ashley Ewearitt, a Counseling and Psychological Services (CAPS) counselor, is embedded in UMSI, schedule an appointment with her by copying and pasting the following url into your browser: <https://bit.ly/Ewearitt-CAPS-UMSI>

For personal concerns, U-M offers a variety of resources, many which are listed on the [Resources for Student Well-being webpage](#). You can also search for additional well-being resources on that website.

Audio and Video Recording Policy

We will be doing audio and video recording of all lecture sessions to enable those who cannot attend class in person on a given day to access the content. These recordings will not be made available publicly. Recordings of all sessions will be available on Canvas only to students registered for this class. As part of your participation in this course, you may be recorded. If you do not wish to be recorded, please contact the professor during the first week of class to discuss alternative arrangements. The camera only picks up the front of the room (instructor and slides), but this may require you to sit in a particular place in the room, outside the cameras' view. Our classroom does not have a ceiling mic that picks up student voices, in addition the instructor's microphone records audio in the room. Students may not copy and share the lecture videos with those not in the class, or upload them to any other online environment (this is a violation of the Federal Education Rights and Privacy Act (FERPA)).

Personal recordings are prohibited except with permission. Students are prohibited from recording/distributing any class activity without written permission from the instructor, except as necessary as part of approved accommodations for students with disabilities. Any approved recordings may only be used for the student's own private use.